

Research, Solution & Implementation Plan Summary

Project: Surplus Food Redistribution Network

Team: Catalyst Alliance

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Proposed Solution: Baki

UN Sustainable Development Goal: Goal 2 – Zero Hunger

Last Updated: 13 August 2026

1. Research Summary

Food insecurity and food waste remain significant challenges across Africa. According to the **Africa Regional Overview of Food Security and Nutrition 2025**, Africa is not on track to achieve Sustainable Development Goal 2 (Zero Hunger) by 2030. More than 300 million people in Africa were undernourished in 2024, while nearly 60% of the population experienced moderate or severe food insecurity. Rising food prices, economic pressures, climate change, and conflict further affect access to nutritious food.

At the same time, significant quantities of food are lost or wasted. The **UNEP Food Waste Index Report 2024** estimated that 1.05 billion tonnes of food were wasted globally in 2022. Food waste occurs across households, food services, retail, and other stages of the food system. This creates an opportunity to recover edible surplus food and redirect it to people and organizations that can use it.

Research into food redistribution identified several barriers that can prevent surplus food from reaching communities in need. These include limited awareness of nearby organizations accepting donations, slow communication between donors and recipients, transportation and coordination challenges, limited cold-chain infrastructure for perishables, and the lack of a centralized system connecting food donors with potential recipients.

The project also analyzed a publicly available operational inventory dataset to demonstrate how data can be used to identify and prioritize surplus food. The analysis identified **2,193 overstocked inventory units**, with Fruits and Vegetables recording the highest surplus and one of the slowest inventory turnover rates. Dairy products also showed high surplus and slow movement. Additional analysis identified products, suppliers, and warehouse locations requiring greater attention.

Because publicly accessible operational retail food-waste datasets for African countries are limited, the inventory dataset was used to demonstrate the analytical approach rather than as direct evidence of African retail waste levels. African reports, statistics, and literature were used to establish the project's

problem context.

2. Proposed Solution – Baki

Baki, meaning “*what's left*” or “*the remainder*” in Swahili, is a lightweight digital surplus food redistribution platform designed to address the coordination gap between food vendors with surplus food and nearby organizations or community members who can use it.

The research findings directly informed Baki's design, particularly the need for faster, more localized coordination between food donors and potential recipients.

Vendors can create surplus food listings by providing information such as food type, quantity, expiry date, and pickup location. They can then choose between two pathways:

- **Donation:** Surplus food can be offered to verified pantries and NGOs.
- **Discounted sale:** Surplus food can be offered to nearby community members at a vendor-set price.

For donations, Baki's matching system identifies suitable nearby organizations based on factors such as **distance, urgency, and available capacity**, and sends the selected organization a pickup notification. The listing can also remain visible on a public live board if the initial match is not collected in time.

For discounted sales, listings are posted on the same public board, allowing nearby community members to discover available surplus food. Community subscribers can also receive notifications about relevant listings.

The platform therefore functions primarily as a **coordination layer rather than a logistics or delivery service**. Its purpose is to make surplus food visible, connect it with potential recipients quickly, and reduce communication delays that can cause edible food to become waste.

3. Implementation Plan

The project will be implemented as a website prototype demonstrating the core Baki concept and user experience. The initial focus is on the African context, particularly Nigeria, South Africa, and Kenya, reflecting the countries represented by the project team.

The implementation will follow an iterative approach consisting of five main phases:

Phase	Focus	Key Output
Planning	Define scope, objectives, roles, and project requirements	Project Charter & Management Plan
Research	Validate the problem, conduct research, and analyze available data	Research Report & Data Analysis Report
Design & Development	Develop Baki's website prototype and technical components	Website Prototype & GitHub Repository
Testing & Refinement	Review functionality, usability, security considerations, and documentation	Refined Prototype & Documentation
Submission	Finalize project materials and prepare the presentation and walkthrough	README, Presentation & Walkthrough Video

The prototype will demonstrate key functionality including surplus food listing, donor and recipient interaction, location-based matching, notifications, pickup coordination, and impact reporting.

The project is intentionally scoped as a prototype. Mobile application development, full deployment, payment integration, and real-time logistics are outside the current project scope.

The implementation will also incorporate cybersecurity considerations and testing before final documentation and submission. Project progress will be monitored through team meetings, milestone reviews, action-item tracking, task-owner updates, and ongoing documentation.

Conclusion

Research indicates that food insecurity and food waste can coexist, creating an opportunity to improve the distribution of existing food resources. The project identified coordination and visibility as important barriers to effective surplus food recovery and used these findings to shape Baki, a technology-enabled platform for connecting surplus food with nearby organizations and communities.

By combining research, inventory analysis, and an implementable digital prototype, Catalyst Alliance aims to demonstrate how technology can support faster surplus food recovery, reduce food waste, and contribute to SDG 2 – Zero Hunger.